IMPERIAL COLLEGE LONDON

DEPARTMENT OF ELECTRICAL AND ELECTRONIC ENGINEERING
EXAMINATIONS 2019

EEE/EIE PART II: MEng, BEng and ACGI

COMMUNICATION SYSTEMS

Corrected Copy

Wednesday, 5 June 10:00 am

Time allowed: 2:00 hours

There are THREE questions on this paper.

Answer ALL questions.
Q1 carries 40% of the marks. Questions 2 and 3 carry equal marks (30% each).

Any special instructions for invigilators and information for candidates are on page 1.

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Second Marker(s) : J.A. Barria

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EXAM QUESTIONS

Information for Students

Fourier Transform Pairs

Pair Number	$x(t)$	$X(f)$
1.	$\Pi\left(\frac{t}{\tau}\right)$	$\tau \operatorname{sinc} \tau f$
2.	$2W \operatorname{sinc} 2Wt$	$\Pi\left(\frac{f}{2W}\right)$
3.	$\Lambda\left(\frac{t}{\tau}\right)$	$\tau \operatorname{sinc}^2 \tau f$
4.	$\exp(-\alpha t)u(t),\ \alpha > 0$	$\frac{1}{\alpha + j2\pi f}$
5.	$t \exp(-\alpha t)u(t),\ \alpha > 0$	$\frac{1}{(\alpha + j2\pi f)^2}$
6.	$\exp(-\alpha\|t\|),\ \alpha > 0$	$\frac{2\alpha}{\alpha^2 + (2\pi f)^2}$
7.	$e^{-\pi(t/\tau)^2}$	$\tau e^{-\pi(f\tau)^2}$
8.	$\delta(t)$	1
9.	1	$\delta(f)$
10.	$\delta(t - t_0)$	$\exp(-j2\pi f t_0)$
11.	$\exp(j2\pi f_0 t)$	$\delta(f - f_0)$
12.	$\cos 2\pi f_0 t$	$\frac{1}{2}\delta(f - f_0) + \frac{1}{2}\delta(f + f_0)$
13.	$\sin 2\pi f_0 t$	$\frac{1}{2j}\delta(f - f_0) - \frac{1}{2j}\delta(f + f_0)$
14.	$u(t)$	$(j2\pi f)^{-1} + \frac{1}{2}\delta(f)$
15.	$\operatorname{sgn} t$	$(j\pi f)^{-1}$
16.	$\frac{1}{\pi t}$	$-j \operatorname{sgn}(f)$
17.	$\hat{x}(t) = \frac{1}{\pi}\int_{-\infty}^{\infty} \frac{x(\lambda)}{t - \lambda} d\lambda$	$-j \operatorname{sgn}(f)X(f)$
18.	$\sum_{m=-\infty}^{\infty} \delta(t - mT_s)$	$f_s \sum_{m=-\infty}^{\infty} \delta(f - mf_s),$ $f_s = T_s^{-1}$

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Useful Relations and Formulas

$$\tan(x+y) = \frac{\tan x + \tan y}{1 - \tan x \tan y}$$

$$\tan(x-y) = \frac{\tan x - \tan y}{1 + \tan x \tan y}$$

$$\tan\left(\frac{\pi}{2} - x\right) = \cot x$$

$$\tan 2x = \frac{2\tan x}{1-\tan^2 x} = \frac{2\cot x}{\cot^2 x - 1} = \frac{2}{\cot x - \tan x}$$

$$\tan\frac{x}{2} = \pm\sqrt{\frac{1-\cos x}{1+\cos x}} = \frac{\sin x}{1+\cos x} = \frac{1-\cos x}{\sin x}$$

$$\sin x \cos y = \frac{1}{2}[\sin(x+y) + \sin(x-y)]$$

$$\cos x \sin y = \frac{1}{2}[\sin(x+y) - \sin(x-y)]$$

$$\sin x \sin y = \frac{1}{2}[\cos(x-y) - \cos(x+y)]$$

$$\cos x \cos y = \frac{1}{2}[\cos(x+y) + \cos(x-y)]$$

$$\sin x + \sin y = 2\sin\frac{x+y}{2}\cos\frac{x-y}{2}$$

$$\sin x - \sin y = 2\cos\frac{x+y}{2}\sin\frac{x-y}{2}$$

$$\cos x + \cos y = 2\cos\frac{x+y}{2}\cos\frac{x-y}{2}$$

$$\cos x - \cos y = -2\sin\frac{x+y}{2}\sin\frac{x-y}{2}$$

Hilbert Transform

Hilbert transform of a signal $g(t)$ is given by

$$g(t) = \frac{1}{\pi}\int_{-\infty}^{+\infty}\frac{g(\tau)}{t-\tau}d\tau$$

Fourier Transform Theorems[a]

Name of Theorem		
1. Superposition (a_1 and a_2 arbitrary constants)	$a_1x_1(t) + a_2x_2(t)$	$a_1X_1(f) + a_2X_2(f)$
2. Time delay	$x(t - t_0)$	$X(f)e^{-j2\pi ft_0}$
3a. Scale change	$x(at)$	$\lvert a\rvert^{-1}X\left(\frac{f}{a}\right)$
b. Time reversal	$x(-t)$	$X(-f) = X*(f)$
4. Duality	$X(t)$	$x(-f)$
5a. Frequency translation	$x(t)e^{j\omega_0 t}$	$X(f - f_0)$
b. Modulation	$x(t)\cos\omega_0 t$	$\frac{1}{2}X(f - f_0) + \frac{1}{2}X(f + f_0)$
6. Differentiation	$\frac{d^n x(t)}{dt^n}$	$(j2\pi f)^n X(f)$
7. Integration	$\int_{-\infty}^{t} x(t')\,dt'$	$(j2\pi f)^{-1}X(f) + \frac{1}{2}X(0)\delta(f)$
8. Convolution	$\int_{-\infty}^{\infty} x_1(t - t')x_2(t')\,dt'$	$X_1(f)X_2(f)$
	$= \int_{-\infty}^{\infty} x_1(t')x_2(t - t')\,dt'$	
9. Multiplication	$x_1(t)x_2(t)$	$\int_{-\infty}^{\infty} X_1(f - f')X_2(f')\,df'$
		$= \int_{-\infty}^{\infty} X_1(f')X_2(f - f')\,df'$

Q Function

$$Q(x) = \frac{1}{\sqrt{2\pi}}\int_x^{\infty} e^{-t^2/2}dt$$

Joint Gaussian density

The joint probability density function (pdf) of two correlated Gaussian random variables X and Y is given by

$$f_{XY}(x,y) = \frac{1}{2\pi\sigma_X\sigma_Y\sqrt{1-\rho^2}}e^{-\frac{1}{2(1-\rho^2)}\left[\frac{(x-\mu_X)^2}{\sigma_X^2} - \frac{(x-\mu_X)(y-\mu_Y)}{\sigma_X\sigma_Y} + \frac{(y-\mu_Y)^2}{\sigma_Y^2}\right]}.$$

where $\mu_X = E[X]$, $\mu_Y = E[Y]$ are the mean values, σ_X and σ_Y are the standard deviation of X and Y, respectively, and ρ is the correlation coefficient defined as

$$\rho = \frac{\mathrm{Cov}(X,Y)}{\sigma_X\sigma_Y}.$$

1. a) i) What is the *threshold effect* in amplitude modulation? Is it observed in coherent demodulation? Why? [3]

ii) Is it possible to employ a non-coherent envelope detector with phase shift keying (PSK)? Explain. [3]

iii) Let $X(t)$ be a white Gaussian noise process. Consider two distinct samples of this process, say $X(t_1)$ and $X(t_2)$, $t_1 \neq t_2$. Explain whether they are correlated, independent, or both. [2]

iv) Random variable X is defined on alphabet $\mathcal{X} = \{1,2,4,8\}$, and its entropy is given by $H(X) = 0.85$ bits. If random variable Y is defined on alphabet $\mathcal{Y} = \{2,4,8,16\}$ with the exact same probability mass function; that is,

$$P(Y = 2i) = P(X = i), \text{ for } i = 1,2,4,8.$$

What is the entropy of Y? [2]

b) Let $X(t)$ and $Y(t)$ be two random processes. State whether each of the following statements are true or false, and discuss your answer (answers without any explanation will not be accepted):

i) If $X(t)$ is strict sense stationary (SSS), then its mean is time-invariant. [2]

ii) If $X(t)$ is wide sense stationary (WSS), then its Hilbert transform is also WSS. [2]

iii) If $X(t)$ is a Gaussian process, then it is white. [2]

iv) If $X(t)$ and $Y(t)$ are independent and WSS, then $X(t)Y(t)$ is also WSS. [3]

c) Consider the signal $m(t) = 3\cos(2\pi f_m t)$ with $f_m = 4$ kHz. Suppose we uniformly quantize $m(t)$ with $L = 256$ levels.

i) Find the maximum quantization error. [2]

ii) Find the signal-to-noise ratio (SNR) in dB for uniform quantizer. [3]

iii) Suppose m(t) is sampled at 25% over the Nyquist rate. Find the bandwidth for the pulse-coded modulation (PCM) signal. [2]

iv) Assume now that we have to sample at a rate that is 75% higher than the Nyquist rate. But we still would like to transmit at or below the PCM bandwidth of part (iii). How many quantization levels should we use? What SNR (in dB) does this result in? [3]

d) Assume that $X(t)$ is a real wide sense stationary (WSS) random process whose power spectral density (PSD) is given as follows:

$$S_X(f) = \frac{1}{\frac{\pi}{2} + 8\pi\left(f - \frac{1}{4}\right)^2} + \frac{1}{\frac{\pi}{2} + 8\pi\left(f + \frac{1}{4}\right)^2}.$$

i) What is the second moment of the random variable $X(1) - X(1 - \frac{1}{\pi})$? (*Hint: Second moment of a random variable Y is given by $E[Y^2]$.*) [5]

ii) What is the second moment of the following random variable?

$$X(t+1) + X(t+3) - X(t+5) - X(t)$$

[6]

2. a) Consider the random process

$$X(t) = A_1 \cos(2\pi f_c t) - A_2 \sin(2\pi f_c t),$$

where $f_c > 0$ is a given fixed frequency, and A_1 and A_2 are independent zero-mean Gaussian random variables with variance σ^2.

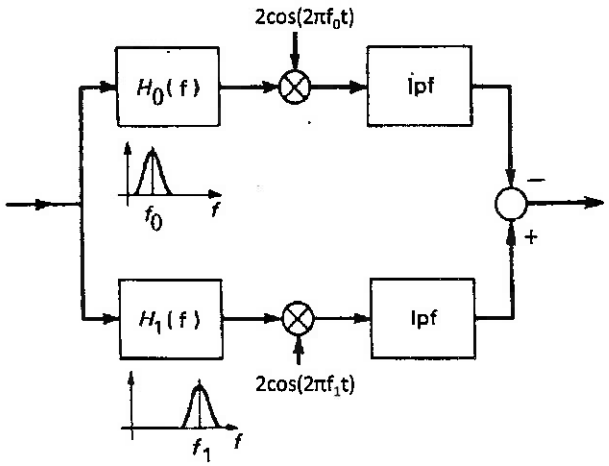
i) Find the mean and autocorrelation function of $X(t)$. [5]

ii) Is $X(t)$ a wide sense stationary (WSS) process? If so, explain why; if not, explain why not. [2]

iii) Find and plot the power spectral density (PSD) of $X(t)$. [4]

iv) Let $Y(t)$ denote the Hilbert transform of $X(t)$. Find the PSD of $Y(t)$. [4]

b) Consider a binary digital communication system with frequency shift keying (FSK) modulation, where bit "0" is transmitted with a rectangular pulse of amplitude $-A$ modulated at frequency f_0, and bit "1" is transmitted with a rectangular pulse of amplitude $+A$ at frequency f_1. The noise power spectral density around frequency f_0 is N_0, while it is $2N_0$ around frequency f_1. The receiver uses the demodulator seen in the figure below, followed by a matched filter and threshold detection.



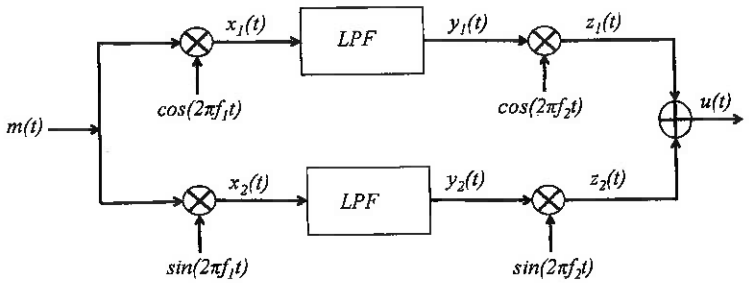
i) What is the probability of error for this communication system in terms of E_b/N_0, where E_b denotes the average energy consumed to transmit each bit of information? [3]

ii) To compensate for the higher noise variance, the transmitter uses a pulse amplitude of $+2A$ to transmit bit "1". What is the probability of error for this communication system in terms of E_b/N_0? Is this smaller than the one obtained in part (i)? [4]

iii) If you can employ binary phase shift keying (PSK) instead of FSK over the same bandwidth resources, which signals would you transmit for each bit to achieve the minimum error probability in terms of E_b/N_0? What is the probability of error? [4]

iv) Can you transmit 2 bits instead of 1 bit over the same channel resources, that is, using the same bandwidth and pulse duration? Explain how you would achieve this. [4]

3. a) Consider the block diagram of a modulator below, which modulates a message waveform $m(t)$ to the output waveform $u(t)$:



Consider modulating a message waveform of $m(t) = A\cos(2\pi ft)$, where $f_L \le f \le f_H$.

i) For $f_1 = (f_L + f_H)/2$, write down the waveforms $x_1(t)$ and $x_2(t)$. [3]

ii) Set the cut-off frequency of the low-pass filters (LPFs) to $f_c = (f_H + 2f_L)/2$, and write down the waveforms $y_1(t)$ and $y_2(t)$ at the output of the LPFs. [4]

iii) Assuming $f_2 >> f_H$ write down the output waveform $u(t) = z_1(t) + z_2(t)$. [3]

iv) What type of modulation scheme is this? What is the carrier frequency? [5]

b) i) For random variables X and Y, let $H(X)$ and $H(Y)$ denote the entropy of X and Y, respectively, while $H(X,Y)$ denote their joint entropy. If X and Y are independent, which one is larger: $H(X)+H(Y)$ or $H(X,Y)$? State the conditions under which $H(X,Y)$ can be larger than $H(X)+H(Y)$. [4]

Let random variables X and Y be defined on alphabets $\mathcal{X} = \{1,2,3\}$ and $\mathcal{Y} = \{a,b,c\}$, respectively. Assume that the joint distribution of X and Y is as given below.

	P(X,Y)	Y = a	Y = b	Y = c
X	1	1/16	3/16	1/16
	2	1/4	1/16	0
	3	0	1/8	1/4

ii) Find $H(X)$ and $H(Y)$. [4]

iii) Find $H(X,Y)$. [4]

iv) Based on the above, can you argue that X and Y are independent? [3]